



Aditya Raj
Computer Science & Engineering
Indian Institute of Technology Bombay

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B.Tech.
Gender: Male
DOB: 27/11/2003

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	
Intermediate	CBSE	Sadar Alam Memorial Secondary School, Bihar Sharif, Nalanda, Bihar	2021	95.20%
Matriculation	CBSE	Ramakrishna Mission Vidyapith, Deoghar, Jharkhand	2019	95.20%

SCHOLASTIC ACHIEVEMENTS

- Secured **29th** rank in the highly competitive **Indian National Mathematics Olympiad (INMO)** (2021)
- Got selected to attend the **International Mathematical Olympiad Training Camp (IMOTC)**, organised by **Homi Bhabha Centre for Science Education (HBCSE-TIFR)** (2021)
- Received **KVPY Fellowship** by **Govt. of India** with **All India Rank 199** among **50,000+** students (2019)
- Achieved **All India Rank 306** in **JEE Advanced** among **over 140,000** eligible candidates (2021)
- Acquired **All India Rank 478** in **JEE Main** among **over 900,000** eligible candidates (2021)

PROFESSIONAL EXPERIENCE

KLA-Tencor | *Algorithm Intern* | *Internship* Summer 2024

- Benchmarked, profiled, and optimized** inspection algorithms for **semiconductor** manufacturing equipment
- Employed **Intel VTune** for **profiling** and identifying performance **hotspots**, leading to **targeted** optimizations
- Optimized **cross-correlation algorithms** by reducing the Region of Interest through **gradient density analysis**

YUN Solutions LLP | *Data Analyst* | *Internship* Summer 2023

- Used **AngelOne's SmartAPI** library to develop an innovative python-based **stock recommendation platform**
- Designed a cutting-edge bot utilizing the **YouTube Data API** and **Whisper AI** technology to efficiently extract transcripts from a curated selection of the most influential stock recommendations on YouTube
- Deployed the powerful combination of **Natural Entity Recognition** and **GPT API** to extract invaluable insights from transcribed data, empowering astute investment decisions grounded in comprehensive analysis

KEY PROJECTS

Cache Hierarchy Optimization | *Prof. Biswabandan Panda* | *Course Project* Spring 2023

- Implemented **cache hierarchies** with varying **L1, L2, LLC sizes and configurations** (inclusive/non-inc/exclusive)
- Examined the impact of different cache **replacement policies** (**LRU, FIFO, LFU and Random**) on performance of different **Graph Algorithms** by plotting **IPC** against inclusivity and cache size for the different algorithms
- Compared the performance of cache hierarchies with a baseline cache hierarchy leveraging **Champsim** simulator

Compiler for SCLP | *Prof. Uday Khedkar* | *Course Project* Spring 2024

- Developed and implemented a compiler for **SCLP (A Language Processor for a Small C-like Language)**
- Used techniques like **lexical, syntax and semantic analysis**, to generate distinct **Intermediate Representations**
- Tackled complexities of handling control flow statements and function calls within the limited C language subset
- Incorporated robust error handling strategies to detect and report syntax and semantic errors during compilation

Sign Language Detection | *Prof. Preethi Jyothi* | *Course Project* Spring 2024

- Developed a **real-time** sign language interpretation system utilizing **Convolutional Neural Network (CNN)** and **Feed-forward Neural Network (FNN)** models, achieving accuracies **98.07%** and **99.68%** respectively
- Utilized **OpenCV** and **MediaPipe** to develop a comprehensive custom dataset for sign language alphabet recognition and successfully implemented **live video processing** with hand segmentation and joint localization

Enhancements to xv6 OS | *Prof. Purushottam Kulkarnii* | *Course Project* Autumn 2023

- Implemented **system calls** for process stats and memory details, enhancing system monitoring capabilities
- Introduced **mmap syscall** with **Page Fault handling** for **Lazy memory allocation**, enhancing memory efficiency
- Introduced **Multithreading** using **clone, join syscalls**, alongside diverse CPU scheduling approaches like **FCFS**

End-to-End encrypted Chat | *Prof. Manoj Prabhakaran* | *Course Project* Spring 2024

- Developed a chat application with **end-to-end encryption** and **authentication**, using **RSA-OAEP** and **RSA-PSS** for asymmetric key exchange, and **AES-CBC** and **HMAC** for message encryption and authentication
- Utilized **PyCryptodome**, and **sockets** to enable **secure communication** between clients and a central server
- Designed a scalable architecture with separate **session keys** for each client, ensuring integrity and confidentiality

OTHER PROJECTS

Rail Planner | Prof. Supratik Chakraborty | Course Project

Autumn 2022

- Designed a remarkably efficient **Rail Planner** through the skillful implementation of several effective **Data Structures** including Dictionaries, Linked List, Binary Search Trees, AVL Trees, Tries, Heaps, & Priority Queues
- Skillfully incorporated some highly efficient algorithms like, **Quick Sort** on Linked List and **KMP** for searching Station Names and Journey Reviews and **Heapify** on Binary Search Trees & AVL Trees
- Implemented **Priority Queues** on reviews with numerical rating at least as large as a user-provided threshold

Image Processing | Prof. Suyash P. Awate | Course Project

Autumn 2022

- Developed a **MATLAB** program to generate representative fruit images and create a comprehensive image database
- Utilized **Principal Component Analysis** to analyze 28x28 handwritten digits from the **MNIST** database

Socket Programming | Prof. Bhaskaran Raman | Course Project

Spring 2023

- Created a powerful file transferring system using socket programming, which enabled efficient communication between a central server and **multiple clients**, allowing **concurrent** and efficient file transfers to be executed
- Streamlined the **file transfer** process by facilitating seamless communication between the server and multiple clients, improving system **efficiency** and enhancing the overall user experience in managing file transfers

Best Offset Prefetcher | Prof. Biswabandan Panda | Course Project

Spring 2023

- Conducted a thorough analysis of **best offset** prefetcher paper, grasping its architecture, strengths, and limitations
- Replicated the prefetcher's **architecture** in C++ and integrated it with the open source simulator **ChampSim**
- Tested it against other prefetchers and evaluated its performance on different workloads and instruction set sizes

SAT Solver | Prof. Ashutosh Gupta | Course Project

Spring 2023

- Developed the encoding scheme to translate the game constraints into a Boolean satisfiability (**SAT**) problem
- Used **z3 Python** library to find the winning moves for the given game by solving the satisfiability problem

Bubble Shooter | Prof. Parag Chaudhary | Course Project

Autumn 2021

- Developed Bubble Shooter using **Simplecpp graphics**, **OOP**, vectors, random bubbles, and dynamic controls
- Included features like **time tracking**, **scoring system**, and **energy levels** for a user-friendly experience

Python Combat | Guide: Prof. Kavi Arya | Course Project

Autumn 2022

- Developed an interactive **python** tutorial with **animations** and **videos**, using **HTML**, **CSS**, and **Javascript**
- Embedded a **python interpreter** to facilitate code submission and provide **evaluation** of code line by line
- Used tools like **CodeMirror** and **HTML5 canvas** to create an immersive coding environment and game arena

TECHNICAL SKILLS

Programming	C++, C, Python, Java, Bash, AWK, Sed, Prolog, MATLAB
Web Development	HTML, CSS, Bootstrap, Javascript, Brython, SQLite, MongoDB
Documentation and Others	L ^A T _E X, Doxygen, Sphinx, gdb, Git

POSITION OF RESPONSIBILITY

Teaching Assistant

Spring 2023

Guide: Prof. Mythili Vutukuru | CS101

IIT Bombay

- Engaged proactively with around **30** junior students, offering guidance and addressing queries, helping in a smooth 4-month **CS101** course which was primarily focused on **introducing C** and its complexities to first year students

EXTRACURRICULARS

Cultural	• Learned to play fipple flute at RamaKrishna Mission Vidyapith, Deoghar (2016-18)
Sports	• Led the Yoga group at RamaKrishna Mission Vidyapith, Deoghar (2018) • Completed one year training programme of NOCS-National Sports Organization (2021-2022)
RC Plane	• Collaborated as part of a four-member team to design and build a radio-controlled plane for a competitive event organized by the Aeromodelling Club, IITB (Autumn 2022)
Bootcamp	• Attended Tinkering Bootcamp , conducted by Learner's Space , and got familiarized with Arduino , Internet of Things (IoT) and RaspberryPi (RPi) (Summer 2022)
Hackathon	• Placed among top 58 teams who qualified in the Limestone challenge organised by Tower Research Capital in collaboration with Finance Club , IITB (Spring 2023)